

Student Answer Sheet: Practical 6-2

Practical Title: Practical 6-2 – Re-Engineer Legacy Codes

Student Information

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Task 1: Purpose of the Code

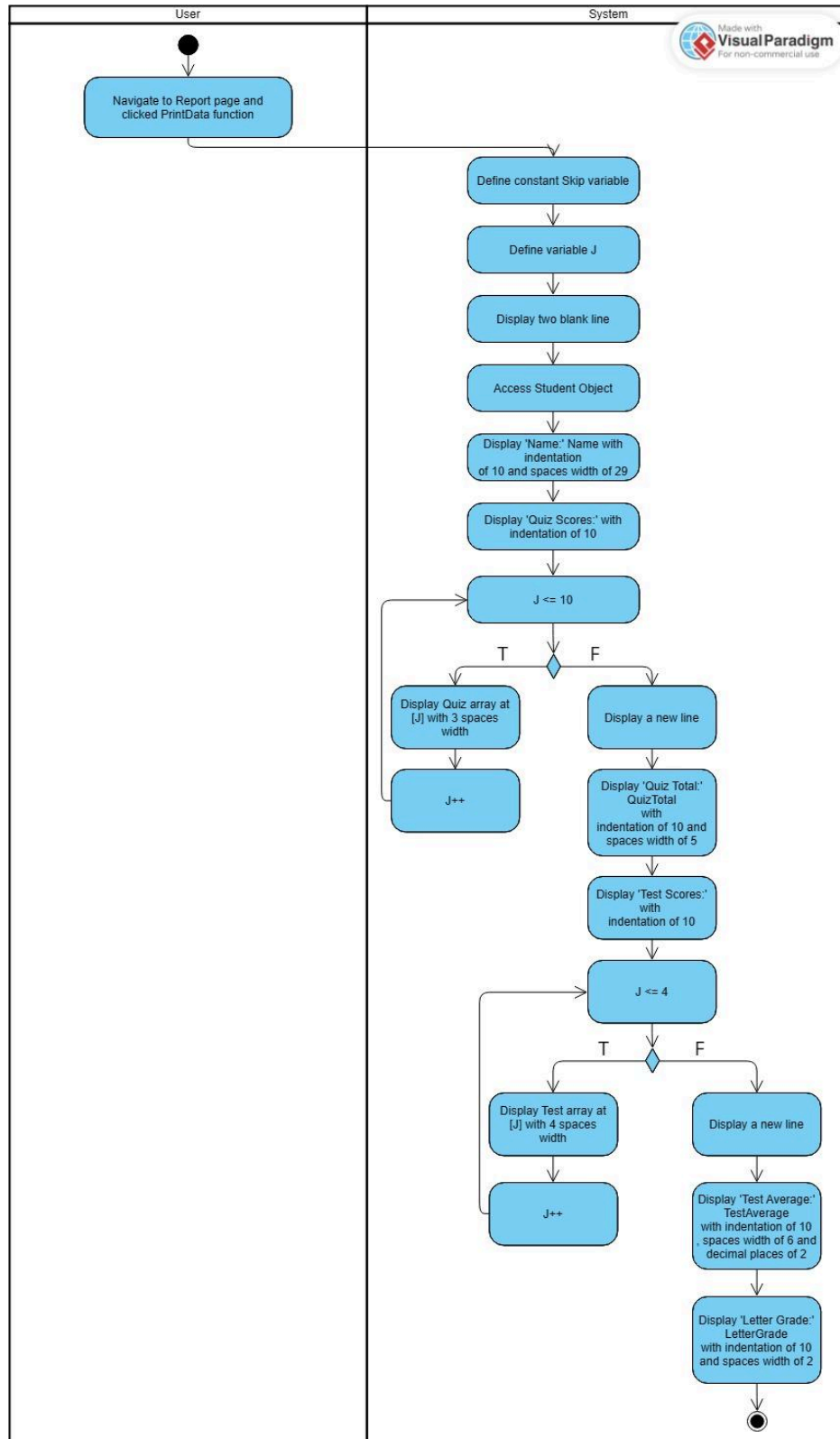
Instructions: Examine the provided legacy Pascal code and explain its primary function and logic in your own words.

Pascal Code	Explanation
PROCEDURE PrintData (Student : StudentRecord);	Define a PrintData function that receive StudentRecord from Student Object
CONST	Define constant variables
Skip = ' ' ;	constant character string Skip used for indentation
VRB	A typo, should be VAR that stands for Variable Section
J : integer;	constant Integer J used as a Loop Counter
BEGIN	Marks the start of executable code
writeln;	Print a new line
writeln;	Print a new line
WITH Student DO	Access Student Object's field
BEGIN	Marks the start of compound statement
writeln (Skip:10, 'Name:', Name:29);	Print new line with indentation of 10 and Name with 29 spaces width
write (Skip:10, 'Quiz Scores:');	Print line with indentation of 10 and 'Quiz Scores' string

<pre> FOR J := 1 TO 10 DO write (Quiz[J]:3); writeln; writeln (Skip:10 ,* 'Quiz Total:', QuizTotal:5); write (Skip:10, 'Test Scores:'); FOR J := 1 TO 4 DO write (Test[J]:4); writeln; writeln (Skip:10, 'Test Average:', TestAverage:6:2); writeln (Skip:10, 'Letter Grade:', LetterGrade:2) END END;</pre>	<pre> for J from 1 to 10 Print line for Quiz array at [J], with 3 spaces width Print a new line Print a new line with indentation of 10 and QuizTotal with 5 spaces width Print line with indentation of 10 and ‘Test Scores’ string for J from 1 to 4 Print line for Test array at [J], with 4 spaces width Print a new line print a new line with indentation of 10 and TestAverage with 6 spaces width and 2 decimal places Print a new line with indentation of 10 and LetterGrade with 2 spaces width End compound statement End executable code</pre>
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Task 2: Activity Diagram

Instructions: Draw or insert the activity diagram representing the execution logic of the PrintData procedure. You must draw it visual paradigm online or other uml tools



Task 3: Java Re-engineering Implementation

Instructions: Paste your re-engineered Java code below. Ensure your program includes the data structures and methods required to reproduce the exact formatted output.

Main.java:

```
public class Main {

    public static void main(String[] args) {

        int[] sampleQuizzes = {85, 90, 78, 92, 88, 95, 80, 82, 89, 91};
        float[] sampleTests = {88.5f, 92.0f, 79.5f, 85.0f};

        Student studentRecord = new Student("Ali", sampleQuizzes, 45, sampleTests, 86.25f, "A");

        printData(studentRecord);

    }

    public static void printData (Student studentRecord) {
        System.out.println("");
        System.out.println("");

        System.out.printf("%10s%29s%n", "", "Name:", studentRecord.getName());
        System.out.printf("%10s", "", "Quiz Scores:");

        int[] quizzes = studentRecord.getQuizScore();
        for (int j = 0; j < quizzes.length; j++) {
            System.out.printf("%3d", quizzes[j]);
        }

        System.out.println("");
        System.out.printf("%10s%5d%n", "", "Quiz Total:", studentRecord.getQuizTotal());
        System.out.printf("%10s", "", "Test Scores:");

        float[] test = studentRecord.getTest();
        for (int j = 0; j < test.length; j++) {
            System.out.printf("%4.1f", test[j]);
        }

        System.out.println("");
        System.out.printf("%10s%-14s%6.2f%n", "", "Test Average:", studentRecord.getTestAverage());
        System.out.printf("%10s%2s%n", "", "Letter Grade:", studentRecord.getLetterGrade());
    }
}
```

Student.java:

```
public class Student {
    private String name;

    private int[] quizScore;
    private int quizTotal;

    private float[] test;
    private float testAverage;

    private String letterGrade;

    public Student(String name, int[] quizScore, int quizTotal, float[] test, float testAverage, String letterGrade) {
        this.name = name;
        this.quizScore = quizScore;
        this.quizTotal = quizTotal;
        this.test = test;
        this.testAverage = testAverage;
        this.letterGrade = letterGrade;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public int[] getQuizScore() {
        return quizScore;
    }

    public void setQuizScore(int[] quizScore) {
        this.quizScore = quizScore;
    }

    public int getQuizTotal() {
        return quizTotal;
    }
}
```

Task 4: Program Output Screenshot

Instructions: Run your Java program and paste a screenshot of the console output below to verify it matches the sample output provided in the prompt.

```
Name: Ali
Quiz Scores: 85 90 78 92 88 95 80 82 89 91
Quiz Total: 45
Test Scores: 88.592.079.585.0
Test Average: 86.25
Letter Grade: A
```